

## SEMESTER S8

### SPEECH AND AUDIO PROCESSING

(Common to CS/CA/CM/CD/CR/AD/CC/CG)

|                                       |                 |                    |                |
|---------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                    | <b>PECST866</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week (L:T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                        | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>         | PECST636        | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To get familiarised with speech processing and audio processing concepts.
2. To equip the student to apply speech processing techniques in finding solutions to day-to-day problems

## SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Speech Production :- Acoustic theory of speech production; Source/Filter model - Pitch, Formant; Spectrogram- Wide and narrow band spectrogram; Discrete model for speech production; Short-Time Speech Analysis; Windowing; STFT; Time domain parameters (Short time energy, short time zero crossing Rate, ACF); Frequency domain parameters - Filter bank analysis; STFT Analysis.                                                                  | <b>9</b>             |
| <b>2</b>          | Mel-frequency cepstral coefficient (MFCC)- Computation; Pitch Estimation ACF/AMDF approaches; Cepstral analysis - Pitch and Formant estimation using cepstral analysis; <i>LPC Analysis</i> - LPC model; Auto correlation method - Levinson Durbin Algorithm                                                                                                                                                                                           | <b>9</b>             |
| <b>3</b>          | Speech Enhancement :- Spectral subtraction and Filtering, Harmonic filtering, Parametric resynthesis; Speech coding - fundamentals, class of coders : Time domain/spectral domain/vocoders, Sub band coding, adaptive transform coding, phase vocoder; Speaker Recognition :- Speaker verification and speaker identification, log-likelihood; Language identification - Implicit and explicit models; Machine learning models in Speaker Recognition. | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                              |          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>4</b> | Signal Processing models of audio perception - Basic anatomy of hearing System, Basilar membrane behaviour; Sound perception - Auditory Filter Banks, Critical Band Structure, Absolute Threshold of Hearing; Masking - Simultaneous Masking, Temporal Masking; Models of speech perception. | <b>9</b> |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written ) | Total     |
|------------|-----------------------------|----------------------------------------|------------------------------------------|-----------|
| <b>5</b>   | <b>15</b>                   | <b>10</b>                              | <b>10</b>                                | <b>40</b> |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                                 | Part B                                                                                                                                                                                                                                                                                | Total     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 subdivisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                              | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|--------------------------------------------------------------|------------------------------------|
| <b>CO1</b>     | To recall various steps in the speech production process     | <b>K2</b>                          |
| <b>CO2</b>     | To summarise various speech processing approaches            | <b>K2</b>                          |
| <b>CO3</b>     | To develop speech-processing applications in various domains | <b>K3</b>                          |
| <b>CO4</b>     | To analyse the speech processing model for audio perception  | <b>K4</b>                          |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   | 2   |     | 2   | 2   |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 3   | 3   | 2   |     |     | 2   |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                                 |                                                       |                              |                         |
|-------------------|-----------------------------------------------------------------|-------------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                        | <b>Name of the Author/s</b>                           | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| <b>1</b>          | Speech Communications: Human & Machine                          | Douglas O'Shaughnessy                                 | IEEE Press                   | 2/e, 1999               |
| <b>2</b>          | Discrete-Time Speech Signal Processing: Principles and Practice | Thomas F. Quatieri                                    | Prentice Hall                | 1/e, 2001               |
| <b>3</b>          | Fundamentals of Speech Recognition                              | Lawrence Rabiner, Biing-Hwang Juang, B. Yegnanarayana | Pearson                      | 1/e, 2008               |

| <b>Reference Books</b> |                                                                                |                             |                              |                         |
|------------------------|--------------------------------------------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                                       | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| <b>1</b>               | Theory and Application of Digital Processing of Speech Signals                 | Rabiner and Schafer         | Prentice Hall                | 1/e, 2010               |
| <b>2</b>               | Speech and Audio Signal Processing: Processing and Perception Speech and Music | Nelson Morgan and Ben Gold  | John Wiley & Sons            | 2/e, 2011               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                |
| <b>1</b>                              | <a href="https://youtu.be/Xjzm7S_kBU?si=j11bk3F7gocYjhfg">https://youtu.be/Xjzm7S_kBU?si=j11bk3F7gocYjhfg</a> |